

COURSE 3 - DATA ENGINEERING 2026

C3-063 - CourierSpark Protected Review

Open only after the full Mini-Lab attempt is saved.

Artifact	C3-063-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

Reviewer expectations

- 42 shipment rows, 42 unique shipment_id values, four unique routes and one missing delivery_minutes source value are visible before transformation.
- R1 has 25 shipment rows and is intentionally hot; diagnosis must distinguish valid business concentration from an invalid key.
- Clean/rejected policy is explicit and reconciles exactly to source.
- route_id dimension uniqueness is proven before join; no join fan-out is tolerated.
- Route metrics and window metric have separately stated grain.
- Plan evidence identifies physical movement/operator choices without claiming local timing proves cluster behavior.
- Parquet output, tests and CLI/runbook are reproducible.

Changed recovery

If one competency failed, repair only that skill and solve a changed shipment/route micro-case before proceeding to Gate A. Completing repair never auto-opens Gate B/C.